

Weekly Progress Report

Refactoring Detection Project - Week 9 progress update for the research supervisor meeting

1. Overview

During Week 9, the CLONE work moved into the final-deliverables stage. The team focused on polishing the remaining UI details, recording professional short and long demo videos, finishing the Overleaf paper, completing documentation, and preparing user-study results for the final report. The Week 9 slides summarize progress in five major areas: demo videos, Overleaf, UI fixes, tests, and website/files.

The short demo reached its final version and was uploaded to OneDrive, while the long demo was prepared and reviewed. The team also continued final edits in Overleaf, updated UI screenshots to IntelliJ light mode, prepared a full updated UI-panel image, and worked toward adding tester feedback graphs to the paper. The website and shared files remained up to date so that the final report, demos, screen captures, README updates, and documentation can be organized in one place.

2. Weekly Progress Summary

Area	Progress Made
Demo videos	Completed and uploaded the final short demo. A longer demo version was uploaded in draft form and prepared for feedback, editing, and finalization.
Overleaf paper	Continued final edits and additions, including updated feature explanations, workflow content, screenshots, and experimental-results material from user testing.
UI fixes	Fixed the size and presentation of the Help button and identified that the custom white-bordered icon should be replaced with IntelliJ's built-in Help icon and a Help tooltip.
Tests / user study	Continued gathering user feedback and planned to convert the form responses into graphs for the final research paper.
Website and files	Kept the project website and OneDrive organized with the latest reports, demo materials, UI screen captures, and documentation files.

3. Meeting Discussion

The Week 9 meeting focused on polishing the final presentation of the CLONE work. Dr. AlOmar gave specific UI feedback about the Help icon, noting that the white bordered square looked more like a button than a contextual help icon. The recommended fix was to use IntelliJ's built-in Help icon so the icon automatically matches the current theme and behaves like a standard IntelliJ toolbar/contextual help element.

The meeting also emphasized final deliverables: professional demo videos, complete documentation, a screen capture in light mode, testing with the 2026 IntelliJ version, user-study results, website updates, README updates, and final paper cleanup. The team was also asked to

address Overleaf comments, proofread the full paper, send user instructions, and reference every figure and table using LaTeX references.

Topic	Decision / Action
Help icon	Replace the custom help image with IntelliJ's built-in Help icon and add a tooltip such as Help.
Demo videos	Create professional short and long demos using meaningful examples instead of placeholder names. The short version should show an end-to-end workflow, and the long version should explain details more fully.
Documentation	Complete full documentation, add a light-mode screen capture, update the README, and update the team website with final resources.
Testing	All team members should test the plugin functionality with the 2026 IntelliJ version and report any issues.
Final paper	Add the complete workflow, address Overleaf comments, proofread the whole paper, reference all figures/tables, and include user-study results.

4. Demo Video Progress

A major Week 9 focus was producing clear and professional demo videos. The short demo was finalized and uploaded, and the long demo was prepared for further review. The demo work was organized around a meaningful Java example rather than placeholder code, so viewers can understand the actual maintenance problem caused by copy-and-paste duplication.

The revised demo structure emphasizes the developer's decision process. Instead of simply touring UI labels, the demo answers four questions: what CLONE found, what the refactoring changes, why the developer should trust the recommendation, and who controls the final decision. The short demo also shows Apply, the updated source code, passing verification, and one control/safety case to make the tool behavior more concrete.

Demo Item	Week 9 Status
Short demo	Final version completed and uploaded to OneDrive.
Long demo	First version uploaded and awaiting feedback; final version still needs to be completed and polished.
Script	Demo narration was prepared around the developer decision story: detect, review, validate, apply, and verify.
Editing focus	Remove long idle processing waits, keep evidence readable, match narration to visible validation states, and close with developer control.

5. Overleaf and Paper Progress

The Overleaf paper continued to move toward final submission. The team added more detail to all sections, updated the images of individual UI sections to light mode, and updated the full UI-panel figure. The paper now needs a stronger Experimental Results section based on the user feedback data collected through the form.

The team also planned to add a complete workflow discussion based on the reference material shared by Dr. AlOmar, especially the workflow ideas from the assigned paper pages. This workflow section should explain how CLONE moves from copy/paste detection to occurrence review, diff inspection, validation evidence, developer action, and final verification.

Paper Area	Required Week 9 Work
Workflow	Add a complete CLONE workflow section explaining the sequence of detection, review, validation, and developer decision.
Experimental results	Report the user-study responses and include graphs showing overall experience and most useful features.
Figures	Use updated light-mode screenshots and reference every figure and table in the text with LaTeX <code>\ref{...}</code> .
Overleaf comments	Address advisor comments and add replies so the paper can be reviewed again.
Proofreading	Proofread the whole paper for clarity, consistency, and correct references.

6. UI Fixes and Final Tool Verification

The main UI fix discussed during Week 9 was the Help button. The current white bordered square around the help icon was considered visually confusing because it looked like a standard button rather than a contextual help icon. The planned improvement is to use IntelliJ's built-in Help icon instead of a custom image, which will allow the icon to match Light, Dark, and New UI themes automatically. A simple tooltip such as Help should also be added.

The team also needs to complete functionality testing using the 2026 IntelliJ version. This verification is important because the final demo, screenshots, README documentation, and user instructions should all reflect the same stable build. Juliana is responsible for creating an updated build, while all team members should test the functionality and report issues.

Item	Reason / Expected Result
Help icon update	Makes the panel follow IntelliJ conventions and prevents users from confusing the help icon with a separate action button.
Tooltip	Adds simple contextual guidance and improves discoverability of the help feature.
Updated build	Provides a stable version for final testing, screenshots, documentation, and demo recording.
IntelliJ 2026 testing	Confirms that the plugin functions correctly in the expected IDE version before final submission.

7. Website, README, and User Instructions

Week 9 also included final documentation work outside the paper. The team website needs to be updated from a weekly tracker into a complete project hub containing a main page, team information, resources, final report, long demo video, and UI screen capture. The GitHub README also needs to explain the relationship between CLONE and AntiCopyPaster, document the CLONE UI panel, and include screenshots of the main panel, occurrence dialog, diff preview, and AI Trust validation area.

User instructions also need to be sent to testers. These instructions should standardize how users install the plugin, configure the model provider, run a small duplicate-code example, understand token requirements, test the panel, and complete the feedback form. Clear instructions help make the user-study results more consistent and easier to interpret.

8. Week 9 Task List

Task	Owner / Notes
Professional demo videos	Create short and long demos with meaningful examples and scripts; short demo finalized, long demo to be finalized.
Full documentation	Everyone contributes; include light-mode screen capture and final feature images.
2026 IntelliJ testing	All team members test the plugin with the 2026 IntelliJ version.

Task	Owner / Notes
Team website update	Add main page, team info, resources, final report, long video, and UI screen capture.
GitHub README update	Document CLONE relationship, UI panel, screenshots, validation policy, and developer controls.
Updated build	Create a stable build for final verification and recording.
User study results	Report collected tester responses and add graphs to the paper.
Paper workflow	Add complete workflow section using the reference paper guidance.
Overleaf comments	Address all comments and add replies for advisor review.
Proofreading and references	Proofread the whole paper and reference every figure/table with \ref{...}.

9. Challenges and Blockers

The largest Week 9 challenge was coordinating many final-deliverable tasks at the same time. The team needed to finalize the UI fixes, prepare a stable build, record and edit two demo videos, finish the Overleaf paper, update the website and README, collect user-study results, and ensure all final visuals were consistent. This required careful organization because changes in the build, screenshots, documentation, and demo videos all depend on each other.

Another challenge was ensuring that the demo and paper make accurate claims. The final demo must match what is visible on screen, avoid long processing pauses, show an end-to-end Apply result, and provide evidence that the project builds and tests pass. The paper must also present validation and user-study results carefully, treating them as evidence rather than guarantees.

10. Immediate Next Steps

- Finalize the long demo video and apply the latest feedback to pacing, narration, readability, and evidence matching.
- Complete the IntelliJ Help icon fix using the built-in Help icon and a Help tooltip.
- Create and test the updated build in the 2026 IntelliJ version.
- Finish gathering user feedback and create graphs for the final paper.
- Add the complete workflow section to Overleaf and address all advisor comments.
- Update the GitHub README with the final CLONE UI documentation and screenshot paths.
- Update the website with the main page, team section, resource page, final report link, long demo, and UI screen capture.
- Proofread the entire paper and make sure every figure and table is referenced with LaTeX references.
- Send standardized user instructions and keep all final files organized on OneDrive and the website.

11. Conclusion

Week 9 focused on finalizing the CLONE deliverables. The team completed the short demo, continued work on the long demo, refined the Help button behavior, updated light-mode UI images, continued final Overleaf edits, prepared user-study results, and kept the website and shared files organized. The remaining work is mainly final polishing: finish the long demo, complete the updated build, test in IntelliJ 2026, report user feedback, revise the paper, update README and website resources, and ensure all figures, tables, and documentation are complete.